

Q7(a)(i)

For the lower spring, the force acting on it is the weight of the lower mass. By Hooke's Law,

$$Mg = ke_2 \text{ or } e_2 = \frac{0.200(9.81)}{24.0} = \mathbf{8.18 \text{ cm}} \quad [1]$$

For the upper spring, $e_1 = \frac{2Mg}{k} = \mathbf{16.4 \text{ cm}}$ [1]

Q7(a)(ii)

The total elastic PE stored in the system is $E_{tot} = \frac{1}{2}ke_1^2 + \frac{1}{2}ke_2^2$ [1]

$$E_{tot} = \frac{1}{2}(24.0)(0.0818^2 + 0.164^2) = \mathbf{0.403 \text{ J}} \quad [1]$$

Q7(a)(iii)

When the system is cut at P, the tension in the upper spring is $2Mg$ while the weight of the upper mass is Mg . By N2L, $2Mg - Mg = Ma$ [1]

Hence the upward acceleration of the upper mass is 9.81 m s^{-2} upwards. [1]

Q7(a)(iv)

The student's use of $v = u + at$ is wrong. [1]

This is because the acceleration of the mass is not constant. [1]

Q7(a)(v)

At the instant the system is cut at P, the upper mass will move upwards with simple harmonic

motion with period given by $T = 2\pi\sqrt{\frac{M}{k}}$ [1]

$$\text{Hence } T = 2\pi\sqrt{\frac{0.200}{24.0}} = 0.5736 \text{ s} \quad [1]$$

$$\approx 0.574 \text{ s}$$

Q7(b)(i)

For rotational equilibrium, Clockwise moment = anti-clockwise moment about pivot [1]

$$M_2 g \times d \sin \theta = M_1 g \times d \sin (90^\circ - \theta)$$

$$M_2 \sin \theta = M_1 \cos \theta \quad [1]$$

$$M_2 / M_1 = 1 / \tan \theta = 1 / \tan 18.4^\circ = 3.006 = \mathbf{3.00} \quad [1]$$

Q7(b)(ii)

$$\begin{aligned} \text{The net gain in KE of the masses is } K_{\text{net}} &= \frac{1}{2}M_1 v^2 + \frac{1}{2}M_2 v^2 \\ &= \frac{1}{2}(0.100 + 0.300)(2.06^2) = 0.849 \text{ J} \end{aligned} \quad [1]$$

From conservation of energy, net loss in GPE = net gain in KE. [1]

Hence net loss in GPE = **0.849 J** [1]

Q7(b)(iii)

$$\text{Rate of work done} = P = (F \sin \theta) v$$

$$P = (M_2 g \sin \theta) v \quad [1]$$

$$P = (0.300 \times 9.81) \sin 18.4^\circ \times 2.06$$

$$\mathbf{P = 1.91 \text{ W}} \quad [1]$$

Q7(b)(iv)

By the Law of Energy Conservation, from the initial orientation to the final orientation,

$$\text{Loss in GPE by } M_2 = \text{Gain in GPE by } M_1 \quad [1]$$

$$M_2 g d \cos \alpha = M_1 g (d + d \sin \alpha)$$

$$3M_1 \cos \alpha = M_1 (1 + \sin \alpha)$$

$$3 \cos \alpha = 1 + \sin \alpha \quad [1]$$

$$Og(a)(i)$$